

INTRODUCTION

Local government units are crucial for keeping peace, order, and harmony in communities by addressing disputes, complaints, and grievances from residents. In the Philippines, the barangay acts as the first level of governance where people usually seek help for neighborhood conflicts, misunderstandings, family disagreements, property issues, and other community problems. The Lupong Tagapamayapa and the Katarungang Pambarangay system require barangays to mediate disputes and encourage friendly settlement before cases move to formal court hearings. Republic Act No. 7160, known as the Local Government Code of 1991, recognizes the Katarungang Pambarangay as a community-based process for resolving conflicts quickly, cheaply, and accessibly at the grassroots level.

The Katarungang Pambarangay system encourages mediation, conciliation, and arbitration as alternatives to lengthy and expensive court processes. It also fosters community involvement in resolving disputes. Under this system, barangay officials handle complaints, prepare notices and summons, conduct mediation and conciliation sessions, document settlement agreements, and keep required records for both barangay operations and compliance with the Department of the Interior and Local Government (DILG). Due to these responsibilities, efficient complaint management, organized documentation, timely scheduling, and clear communication are vital to ensure fair, systematic, and transparent handling of disputes.

Despite the significance of the Katarungang Pambarangay system, many barangays still use traditional and manual methods to address complaints and mediation. Complaint records are often written by hand in blotter books, stored in physical folders, or shared through verbal communication and informal messaging apps. Hearing schedules and mediation activities are usually tracked with handwritten logs and notebooks. Official documents such as summons, notices, certifications, and settlement forms are often produced through repeated handwriting or manual entry. While these traditional methods still work, they become harder to manage as the volume of complaints and mediation efforts grows.

Continuing to rely on manual processes creates several problems for barangay dispute resolution. Records may become incomplete, duplicated, lost, or hard to find because of disorganized filing

and paper-based documentation. Keeping track of case progress, hearing schedules, and mediation activities can also take a lot of time, especially when there are many complaints. Additionally, manually preparing reports and Katarungang Pambarangay forms increases the administrative workload and repetitive tasks. Delays in communication, missed hearings, and inconsistent documentation can further hinder the timely resolution of community disputes and grievances

The lack of a centralized records management system also makes it hard for barangay personnel to organize historical records, monitor recurring complaints, and track case progress efficiently. Residents may struggle to follow up on complaints due to limited access and poor monitoring systems. Moreover, maintaining confidentiality and accountability is trickier when sensitive records are handled manually without structured access and organized digital recordkeeping.

This study aims to address the pressing need for a more organized and efficient complaint management system within barangay dispute resolution operations. While digital governance initiatives are expanding in the Philippines, many existing systems focus mainly on broader administrative services and rarely meet the specific needs of the Lupon Tagapamayapa. Current barangay processes still require significant manual work to document complaints, schedule mediations, prepare Katarungang Pambarangay forms, monitor case progress, and generate reports.

To tackle these issues, this research proposes a web-based Lupon Tagapamayapa Concern Management System designed to enhance complaint reporting, case monitoring, documentation, scheduling, and records management in the barangay. The proposed system aims to offer a centralized digital platform to manage complaint records, mediation activities, hearing schedules, settlement agreements, and other Katarungang Pambarangay processes more effectively. The system will also facilitate document generation, searchable records management, monitoring tools, and role-based access control to improve efficiency, transparency, accountability, and organization in barangay dispute resolution services.

This study also seeks to support gradual digital transformation in barangay governance by providing a practical solution that aligns with current Katarungang Pambarangay procedures. Instead of completely replacing existing processes, the proposed system intends to supplement barangay operations by improving records management, communication, monitoring, and access

to complaint-related information. Through this approach, the study aims to show how digital technology can enhance barangay dispute resolution services and enable more effective community-based grievance management.

Project Context

Barangay Tumana is one of the barangays in Marikina City, consisting of 70 neighborhood associations and serving a large residential population and household community. Through the Lupong Tagapamayapa, the barangay is responsible for handling complaints, facilitating mediation and conciliation proceedings, preparing Katarungang Pambarangay documents, and maintaining case records. Records from the barangay indicate that a total of 325 cases were filed in 2024 and 214 cases were filed in 2025, highlighting the continuing demand for effective dispute resolution, documentation, and case monitoring services. Effective management of these activities requires organized documentation, systematic monitoring, and efficient record keeping.

Based on interviews conducted with personnel of the Lupong Tagapamayapa of Barangay Tumana, complaint records, case documents, and hearing schedules are still managed primarily through physical folders, printed forms, and manual documentation processes. As the number of cases continues to increase, barangay personnel experience difficulties in organizing records, retrieving archived documents, monitoring case progress, and preparing reports and required KP documents.

The interviews also revealed the need for a centralized system capable of storing resident profiles, complaint histories, case statuses, mediation schedules, and related documents. Existing manual processes make it challenging to track recurring disputes, access previous records, and monitor ongoing cases efficiently. Additionally, the preparation of summons, notices, agreements, certifications, and other KP forms contributes significantly to the administrative workload of the Lupon personnel.

Although digital transformation initiatives continue to be promoted in the public sector, many barangays still lack systems specifically designed for Katarungang Pambarangay operations. Existing solutions often do not adequately support case monitoring, mediation scheduling, document generation, records management, and reporting requirements within the Lupon

Tagapamayapa.

In response to these challenges, this study proposes the development of the Automated Grievance Assistance Platform (AGAP): A Web-Based Local Concern and Resolution Management System. The proposed system aims to provide a centralized platform for complaint documentation, resident profiling, case monitoring, mediation scheduling, KP document generation, records management, and reporting. Through the digitalization of these processes, the system seeks to improve efficiency, organization, and accountability within the Lupong Tagapamayapa of Barangay Tumana.

After analyzing the gathered data, the researchers identified the following specific problems:

- **How to implement a Login and Role-Based Access Control (RBAC) Module that ensures secure and properly restricted access to the Barangay Case Management System?**

In the current manual setup, case records are handled through shared physical documents such as blotters and logbooks accessed by multiple barangay personnel. This setup does not clearly define user roles or responsibilities in encoding, updating, and validating case information, resulting in unclear accountability in case handling tasks.

- **How to develop a Dashboard, Monitoring, and Reporting System that allows real-time tracking of case progress and operational activities?**

The barangay currently relies on a physical bulletin board or matrix to track the summary of their docketed, settled, or dismissed cases. Manually compiling these statistics for the mandatory quarterly DILG submission and the Lupong Tagapamayapa is highly inefficient and time-consuming.

- **How to design a Search, Retrieval, and Record Organization system that enables efficient storage and quick access to complaint records and historical data?**

Organizing and retrieving profiles is a major challenge, especially since many residents are

tenants who move frequently. Without a searchable digital database, it is difficult to quickly see if a person is a repeated offender or if they have previous unsettled cases.

- **How to implement a centralized Case Management System that allows barangay officials to properly record, organize, and manage grievance cases, including complaints, mediation activities, and resolutions?**

The barangay relies on disjointed manual methods such as handwritten blotters, physical folders to record docketed complaints. Managing a high volume of disputes manually makes it tedious to track the progress of mediation and increases the risk of disorganized or misplaced case files.

- **How to design a Lupon Assignment and User Management System that ensures secure access and proper role-based control for barangay officials and users?**

Currently, barangay records are handled manually without structured digital access, making it difficult to maintain the confidentiality of sensitive dispute records. Without proper role-based control and secure authentication, there is no reliable way to safely assign specific cases to lupon members or restrict access to confidential Karatungang Pambarangay documents to authorized personnel only.

- **How to implement a Scheduling and Hearing Management System that supports mediation processes and organizes hearing schedules effectively?**

Hearing schedules are written manually on the summon and are recorded through a paper-based attendance form. Disputing parties sometimes fail to follow the exact time, causing missed hearing and long waiting periods. Managing the strict 15-day legal deadlines for mediation and conciliation manually is complicated and requires constant monitoring.

- **How to automate document generation, including Katarungang Pambarangay forms and reports required in dispute resolution?**

The Katarungang Pambarangay process requires the preparation of numerous official documents. Preparing these documents manually is time-consuming and increases the risk

of delays in serving notices within the strictly prescribed working day.

- **How to incorporate Location and Proof-of-Service features that enable accurate reporting, verification of incidents, and tracking of on-site actions?**

Lupon members must physically travel to deliver the summons or monitor compliance with settlement agreements. Without a proper location-tracking system, it is difficult to officially verify proof of service, as well as to visualize and pinpoint areas in the barangay where specific issues repeatedly occur.

- **How to develop AI-assisted features that support complaint submission and documentation processes?**

Residents report grievances; they often provide unstructured or emotional verbal accounts. This creates a problem for the lupon secretary who face heavy administrative tasks of manually extracting the essential facts (who, what, when, where, why, and how) from these stories to properly format them into formal blotters or official minutes

Purpose and Description

The purpose of this study is to develop the Automated Grievance Assistance Platform (AGAP): A Web-Based Local Concern and Resolution Management System for the Lupon Tagapamayapa of Barangay Tumana. The proposed system aims to improve the management of complaints, mediation activities, case records, and Katarungang Pambarangay documents through a centralized digital platform.

The system will enable authorized barangay personnel to record and monitor cases, manage mediation schedules and hearings, generate required KP forms and documents, maintain resident and case records, and produce reports needed for operational and administrative purposes. It will also provide search and monitoring tools to facilitate faster retrieval of records and more efficient case tracking.

By digitalizing key Lupon Tagapamayapa processes, the proposed system seeks to reduce manual paperwork, improve record organization, enhance monitoring and reporting capabilities, and

support more efficient dispute resolution services within the barangay. The project focuses specifically on Katarungang Pambarangay operations, providing a specialized solution tailored to the needs of barangay-level complaint and case management.

Objectives

General Objective

To develop a web-based Lupon Tagapamayapa Concern Management System that allows barangay residents to submit and track complaints while enabling barangay officials to efficiently manage grievance cases, mediation processes, and records.

Specific Objectives

- **To design, develop, and implement a Login and Role-Based Access Control (RBAC) Module that ensures secure and properly restricted access to the Barangay Case Management System.**

To address this issue, a centralized Case Management System with a Login and Role-Based Access Control (RBAC) module can be implemented to assign specific roles and permissions to barangay personnel. This ensures that each user has clearly defined access to functions such as encoding complaints, updating case progress, and managing records, thereby improving accountability, workflow organization, and proper case handling.

- **To design, develop, and implement a Dashboard, Monitoring, and Reporting System that enables barangay officials to view, track, and manage grievance cases in real time.**

The dashboard will display an overview of complaint statuses, mediation activities, scheduled hearings, and case progress. It allows administrators to monitor system activity, prioritize cases, and make informed decisions efficiently.

- **To design, develop, and implement a Search, Retrieval, and Record Organization System that enables efficient access, filtering, and management of complaint records and historical case data.**

This module will organize complaint records into a digital database with searchable and filterable features. It allows barangay officials to quickly retrieve past records, track recurring complaints, and access historical case information, improving decision-making and administrative efficiency.

- **To design, develop, and implement a Case Management System that enables the structured recording, categorization, and management of grievance cases, including complaints, mediation processes, and resolutions.**

This system will serve as the core module where all complaint information, mediation activities, and settlement outcomes are digitally recorded and organized. It will ensure complete and consistent documentation, reduce record loss, and enable efficient tracking of case progress throughout the dispute resolution process.

- **To design, develop, and implement a Lupon Assignment and User Management System that ensures secure access, role-based permissions, and proper assignment of cases to barangay officials and Lupon members.**

This feature will provide authentication and role-based access control, ensuring that only authorized users can access sensitive complaint and mediation records. It will also facilitate proper assignment of cases to Lupon members, Pangkat officers, and staff, ensuring accountability and organized workflow in dispute resolution processes.

- **To design, develop, and implement a Scheduling and Hearing Management System that facilitates hearing scheduling, notification of involved parties, and management of mediation sessions under Katarungang Pambarangay.**

This feature will provide a structured scheduling tool that organizes mediation sessions, tracks hearing dates, and prevents missed or conflicting schedules. It supports efficient coordination of dispute resolution activities and ensures timely processing of cases.

- **To design, develop, and implement an Automated Document and KP Form Generation System that produces official documents, reports, and settlement forms required for barangay dispute resolution processes.**

This feature will automate the creation of summons, notices, settlement agreements, certificates, and other Katarungang Pambarangay forms. It reduces repetitive manual work, ensures document consistency, and improves compliance with required documentation standards.

- **To design, develop, and implement Location and Proof-of-Service Features that allow users to attach location data and enable responders to document on-site actions and verification results.**

This feature will allow residents and staff to specify the exact location of incidents and enable responders to record field activities. It supports better verification, faster response coordination, and improved accuracy in handling both complaints and on-site incidents.

- **To design, develop, and implement AI-assisted features, including a chatbot for guided complaint submission and an AI-based narrative generator for report documentation.**

These AI-powered tools will guide residents through the complaint submission process by suggesting appropriate categories and improving clarity. Additionally, the narrative generator will assist barangay staff in converting structured data into formal and organized documentation, reducing administrative workload.

Rationale

Barangay Residents

Barangay residents are among the primary beneficiaries of the proposed Lupon Tagapamayapa Web-Based System. The system provides a more organized and structured way of filing grievances and complaints, reducing reliance on verbal reports and informal communication channels. Through a centralized platform, residents can ensure that their complaints are properly recorded and forwarded to the appropriate barangay personnel for processing. By improving the documentation and management of complaints, the system helps promote a more efficient and responsive grievance handling process within the barangay.

Lupon Tagapamayapa and Barangay Personnel

The proposed system directly benefits the Lupon Tagapamayapa, Lupon Secretary, and other authorized barangay personnel by improving the management of grievance and complaint records. Through centralized digital records, automated document generation, case tracking, scheduling, and reporting tools, the system reduces the difficulties associated with manual record keeping and paper-based documentation. It also facilitates faster retrieval of case information, improves monitoring of complaint status and hearing schedules, and supports the preparation of reports required for administrative purposes. As a result, the system enhances operational efficiency and helps maintain more organized and reliable records.

Barangay Administration and Local Government

The barangay administration may benefit from a digital system that supports the modernization of grievance and complaint management processes. By improving record organization, monitoring, and reporting, the system can assist barangay officials in managing community concerns more effectively. The study also supports broader government initiatives aimed at promoting the use of information technology in public service delivery and local governance. Furthermore, the proposed system may serve as a reference for other barangays seeking to improve their grievance management processes through digital solutions.

Future Researchers and IT Students

This study may serve as a useful reference for future researchers and Information Technology students interested in e-governance, information systems, and digital transformation in local government units. It provides insights into the development and implementation of a grievance and complaint management system within a barangay setting. The study may also contribute to future research on records management, complaint monitoring systems, workflow automation, and the application of technology to improve administrative processes in local governance.

Scope and Limitations

Scope

The study focuses on the development of a web-based Lupon Tagapamayapa Concern Management System for a selected barangay in Marikina City. The system will be accessible to four main user groups: administrators, Lupon clerk, summon servers, and barangay staff (Lupon personnel).

1. Administrator Scope

The Administrator, composed of the Barangay Captain and Barangay Secretary, is granted full access to all system modules. They oversee the entire Katarungang Pambarangay process and ensure proper implementation of procedures and documentation.

The system will allow administrators to:

- **Login and RBAC Module**
 - Manage user accounts, roles, and system access
- **Dashboard, Monitoring, and Reporting Module**
 - Monitor real-time case status and KP activities
 - Generate reports for DILG compliance
- **Search, Retrieval, and Records Management Module**
 - Access complaint records, resident profiles, and case histories
- **Case Management Module**
 - Manage and oversee all grievance cases
 - Monitor mediation, conciliation, and escalation

- **Lupon Assignment and User Management Module**
 - Assign cases to lupon members and personnel
- **Scheduling and Hearing Management Module**
 - Approve and manage hearing schedules
- **Automated Document and KP Form Generation**
 - Generate and approve KP documents such as summons, notices, settlement agreements, and Certificate to File Action (CFA)
- **Location and Proof-of-Service Module**
 - Verify incident locations and service reports
- **AI-Assisted Features**
 - Utilize AI-assisted tools for structured documentation and report generation

2. Lupon Clerk Scope

The Lupon Clerk supports documentation, scheduling, and case processing within the Katarungang Pambarangay system.

The system will allow the Lupon Clerk to:

- **Login and RBAC Module**
 - Log in, log out, and change password
- **Dashboard, Monitoring, and Reporting System**
 - Monitor case progress and assist in report preparation
- **Search, Retrieval, and Records Management**

- Retrieve and manage complaint records and case histories
- **Case Management System**
 - Encode and update complaint records, mediation details, and resolution outcomes
 - Maintain organized and complete case documentation
- **Scheduling and Hearing Management System**
 - Prepare and manage hearing and mediation schedules
 - Monitor deadlines and attendance
 - Receive notifications regarding scheduling updates
- **Automated Document and KP Form Generation**
 - Prepare and generate KP documents such as summons, notices, and agreements
- **Location and Proof-of-Service Features**
 - View location details and verification reports
- **AI-Assisted Features**
 - Use AI tools to assist in generating formal narratives and documentation

3. Summon Server Scope

The Summon Server is responsible for delivering official documents and conducting field verification.

The system will allow Summon Servers to:

- **Login and RBAC Module**
 - Log in, log out, and change password

- **Dashboard, Monitoring, and Reporting System**
 - View updates for assigned cases only
- **Location and Proof-of-Service Features**
 - Access incident locations and involved parties
 - Record and submit proof of service for summons and notices
 - Provide verification updates based on field observations

4. Lupon Members Scope

Lupon members will use the system to report and monitor community concerns.

The system will allow Lupon Members to:

- **Login and RBAC Module**
 - Log in, log out, and change password
- **Dashboard, Monitoring, and Reporting System**
 - View information on assigned cases only
- **Case Management System**
 - View assigned cases and relevant complaint information
 - Access case details necessary for mediation and conciliation
- **Scheduling and Hearing Management System**
 - View assigned mediation schedules and hearing calendar
 - Receive notifications regarding hearing dates and schedule changes

Limitations

Despite the system's capabilities, several limitations exist for each user group.

- The system will not replace government response services and will not integrate directly with external emergency response agencies (such as police, fire departments, or hospitals); therefore, escalation and coordination will still require manual action by barangay officials.
- The system is designed primarily for web browsers and may not provide full functionality or optimal performance on older mobile devices or outdated browsers.
- The implementation is limited to a selected barangay in Marikina City
- The AI does not perform predictive analytics or legal decision-making

METHODOLOGY

Technical Background

Development Approach

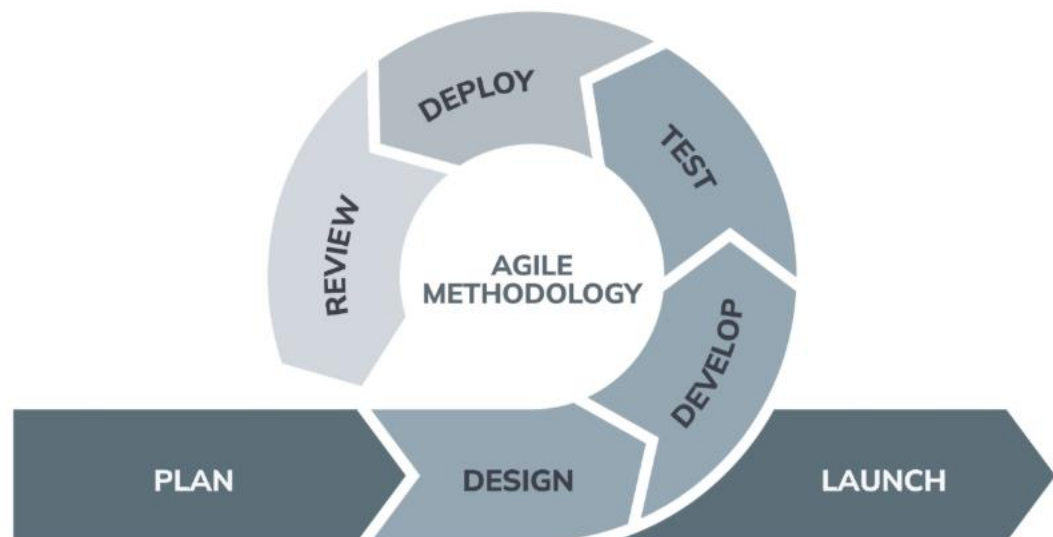


Figure 1. Agile Methodology

The Lupon Tagapamayapa Web-Based System will be developed using an Agile Software Development methodology following a modified Scrum-based approach. This development approach was chosen because it is flexible and appropriate for systems with changing requirements. The development process will be progressive, with phases such as planning, design, implementation, testing, and assessment. Each iteration will result in working system components, which will be examined and refined based on feedback from local stakeholders. This strategy aims to ensure that important modules, such as complaint submission, emergency reporting, and notification systems are designed efficiently while being practical, user-centered, and responsive to barangay operations.

Technologies to be Used

The Lupon Tagapamayapa Web-Based System will be developed using a web-based technology stack selected for its practicality, accessibility, and suitability for barangay-level deployment. The system technologies are chosen based on the proponents' technical capabilities and the operational requirements of local government offices.

Front-End Technologies

The system's user interface will be developed using HTML, CSS, and JavaScript. HTML will be used to structure the web pages, CSS will manage layout and visual presentation, and JavaScript will support client-side interactivity and form validation. Tailwind CSS, loaded via a Content Delivery Network (CDN), will be used to facilitate responsive and consistent interface design without requiring a build process.

An interactive map feature will be implemented using Leaflet.js, an open-source JavaScript mapping library that utilizes free OpenStreetMap tiles and does not require an API key. This will allow residents to indicate the location of reported incidents accurately.

Back-End Technologies

The back-end of the system will be implemented in PHP (version 8.2 or later), without the use of a framework. PHP will handle server-side tasks such as user authentication, session management, form handling, complaint processing, notification logic, and communication with external application programming interfaces (APIs). To improve security and avoid SQL injection, database interactions will be carried out using PHP Data Objects (PDO) and prepared statements.

Database

MySQL will be used as the relational database management system. It will store system data such as user accounts, complaints, emergency reports, assistance requests, notification logs, credibility scores, and audit trail logs. MySQL's compatibility with PHP ensures efficient data storage and retrieval.

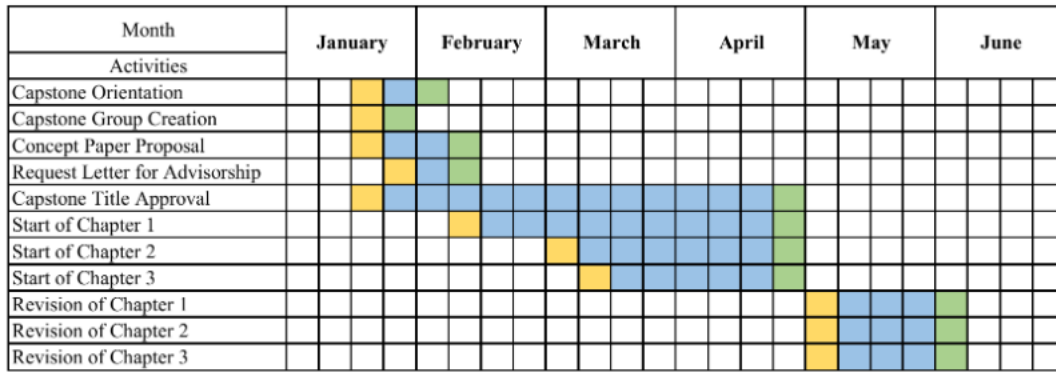
Development and Deployment Environment

During development, Laragon will serve as the local development environment, hosting PHP, MySQL, Apache, and a database management tool for testing. While development takes place locally, the system is meant to be deployed on a web hosting or cloud-based server once completed, allowing for internet access and centralized data storage.

AI Integration

The system will integrate third-party artificial intelligence services through the Google Gemini API to support an AI-powered chatbot and an AI-assisted narrative generator. These features are meant to assist citizens in submitting complaints and to enable barangay officials create structured documentation. AI functionality will solely act as a support tool, with all decisions and validations remaining under human administrative supervision.

Calendar of Activities



Figure

2. Calendar of Activities

Resources

The system will require the following software and hardware resources to support the planned development, testing, and deployment of the Automated Grievance Assistance Platform (AGAP).

Hardware

Hardware Component	Specification
Processor	Multi-core processor equivalent to Intel Core i5 or higher
Memory (RAM)	8 GB RAM or higher
Storage	256 GB SSD or higher
Computer Unit	Laptop or desktop computer
Input Devices	Mouse and keyboard
Internet Connection	Stable internet connection (minimum of 10 Mbps)

Software

The proponents will develop a web-based system to support the reporting and management of community concerns at the barangay level. Barangay staff, and administrators will all have access to the system's various modules. The platform is intended to simplify the reporting process by allowing citizens to easily file complaints and track their status, thereby eliminating the constraints of manual and paper-based reporting systems.

The system front end will be developed using HTML, CSS, and JavaScript, while PHP will be used as the server-side scripting language to handle system logic, authentication, and data processing. MySQL will serve as the database management system, and Laragon will be used as the local development environment during the development and testing phases.

Internet access will be required to operate the web-based platform. The Web-Based System will be accessible through a standard internet connection using modern web browsers during both local testing and planned online deployment.

The system will be developed on a Windows-based operating system (Windows 10 or later). Visual Studio Code (Version 1.88 or later) will be used as the primary code editor for the development of the system due to its support for multiple programming languages and development productivity features.

HTML5 will be used to define and structure all system interfaces, including complaint submission forms, dashboards, notification panels, and account management pages. CSS3, together with Tailwind CSS loaded via a Content Delivery Network, will be used to manage the layout and visual design of the system's web pages. JavaScript (ECMAScript 6 or later) will be used to implement client-side interactivity such as form validation, dynamic content updates, and map-based features.

All back-end system functionalities will be handled using PHP version 8.2 or later. PHP will manage user authentication, session handling, form processing, database interactions, credibility score computation, and notification logic. Database operations will be implemented using the PDO extension with prepared statements to enhance security.

MySQL version 8.4 or later will be used as the relational database management system to store user accounts, complaints, emergency reports, assistance requests, notification records, and credibility scores. MySQL's integration with PHP ensures reliable and efficient data management.

Laragon will be used as a lightweight local development environment that provides a pre-configured setup for PHP, MySQL, and Apache, allowing development and testing without complex configuration. Database management tools included in the environment will support efficient testing and maintenance.

Leaflet.js will be used to implement the system's interactive map feature. This open-source library utilizes free OpenStreetMap tiles and does not require an API key, allowing residents to pin the location of reported incidents accurately and provide barangay officials with geographic context.

The system's AI-assisted features, such as a chatbot to help residents submit complaints and a narrative generator to help administrators create structured documentation, will be supported by integration with the Google Gemini API. AI functions will solely serve as support tools, with all validations and choices falling under administrative management.

Modern web browsers such as Google Chrome or Mozilla Firefox will be used to access and test the system to ensure compatibility, functionality, and responsive design across standard user environments.

Current State of Technologies

Web Technology

Web technology serves as the foundation of the system. The platform is implemented as a web-based application using HTML, CSS, JavaScript, and PHP, allowing users to access the system through standard web browsers without requiring installation. This approach ensures accessibility for barangay residents, staff, and administrators across different devices.

The system provides structured interfaces such as complaint submission forms, administrative dashboards, and real-time status tracking modules. Compared to traditional manual reporting, this web-based setup enables residents to submit Emergency Reports, Grievance Reports, and Assistance Requests remotely, reducing the need for physical visits to the barangay office and improving service accessibility.

Network Technology

Network technology enables communication between system users and the platform through internet connectivity. Using Wi-Fi or mobile data networks, the system supports real-time data transmission, allowing complaints, updates, and notifications to be processed instantly.

This connectivity is essential for time-sensitive operations such as emergency reporting and status updates. Barangay officials can get and reply to submitted reports instantly, while people are notified of developments, which improves coordination and reduces delays as compared to manual communication.

Cloud Computing

Cloud computing plays a role in the deployment phase of the AGAP system. While development and testing are conducted locally using the Laragon environment, the system is intended to be deployed on a web hosting or cloud-based server to enable centralized data storage and remote accessibility.

Through cloud-based deployment, all user accounts, complaints, assistance requests, and report statuses are stored in a centralized database, ensuring data consistency and availability. This setup allows authorized barangay personnel to access system records anytime and supports scalability as the number of users and reports increases.

AI Technology

Artificial Intelligence (AI) is integrated into the system to enhance usability and administrative efficiency. The system utilizes the Google Gemini API to power two key features: an AI-assisted chatbot and a narrative generation tool.

The chatbot assists residents in submitting complaints by guiding them through the reporting process and suggesting appropriate categories and urgency levels. Meanwhile, the AI-assisted narrative generator helps barangay staff convert structured complaint data into clear and organized report summaries.

These AI functions serve solely as assistance aids. All validations, prioritizing decisions, and final actions are still under the supervision of barangay officials, ensuring that human judgment is maintained in the decision-making process.

GPS (Global Positioning System)

GPS technology will be used to enable location-based reporting. Residents will be able to use an interactive map to mark the location of events or concerns, making it easier for barangay officials and tanods to pinpoint affected areas. This functionality is especially useful for emergency circumstances and community-based support requests.

High-Level Programming Language

The system is developed using high-level web programming technologies that support system functionality and maintainability. PHP is used for server-side processing, handling core operations such as authentication, complaint processing, notification logic, and AI integration. JavaScript is used for client-side interactivity, enabling dynamic form validation and responsive user interfaces.

MySQL serves as the relational database management system, storing all system data including user accounts, complaints, emergency reports, assistance requests, notification logs, and credibility scores. The integration of these technologies ensures efficient data processing, reliable storage, and scalable system performance suitable for barangay-level deployment.

Timeliness

This study is timely due to the Philippine government's ongoing digital transformation initiatives. The National Economic and Development Authority Philippine Development Plan 2023 to 2028 emphasizes the modernization of public service delivery through

digitalization. Related foreign studies likewise highlight the importance of adopting information and communication technologies to enhance governance efficiency and citizen participation.

As communities increasingly rely on digital platforms for communication and services, barangays are encouraged to transition from traditional manual procedures to structured digital systems. The proposed integration of web-based reporting, location-based tagging, centralized data storage, and AI-assisted support reflects current technological approaches discussed in recent literature on smart and digital governance systems.

Problem It Solves

The proposed system addresses several challenges identified in the current grievance and complaint management practices of barangays, including:

- Manual and fragmented complaint reporting procedures
- Lack of a structured system for recording, categorizing, and monitoring complaints
- Time-consuming retrieval of complaint information and case histories
- Limited ability to monitor complaint status and resolution progress
- Challenges in identifying complaint trends and recurring issues within the community
- Administrative difficulties in preparing reports and maintaining organized records

Through centralized digital records, complaint tracking, automated documentation, scheduling, and reporting tools, the proposed system aims to improve the efficiency, organization, monitoring, and management of grievance and complaint handling processes within the barangay.

Contribution to Knowledge

This research contributes to the field of e-governance and Information Technology by examining the application of digital grievance and complaint management within the Lupon Tagapamayapa at the barangay level. While many existing studies focus on complaint management systems implemented at municipal, city, or national levels, this study extends the literature by:

- Applying digital governance concepts within the Katarungang Pambarangay system
- Supporting the digitalization of grievance and complaint documentation, scheduling, and records management processes
- Integrating AI-assisted summarization to support administrative documentation and record preparation
- Demonstrating the use of centralized digital records for monitoring complaint status, case progress, and reporting requirements for LTIA

The study bridges theoretical concepts from the related literature with practical system implementation through the integration of web-based technologies, cloud-based record management, and administrative automation tools. Furthermore, it provides insights into how digital solutions can improve the efficiency, organization, and reliability of grievance and complaint management within the Lupon Tagapamayapa.

Possible Implications

If implemented successfully, the proposed system may:

- Improve the efficiency of grievance and complaint handling processes within the barangay
- Enhance the organization, storage, and retrieval of complaint records and related documents

- Strengthen monitoring and tracking of complaint status, schedules, and case progress
- Reduce administrative workload associated with manual documentation and report preparation
- Improve the accuracy and reliability of grievance and complaint records
- Support data-driven decision-making through centralized records and complaint statistics
- Contribute to ongoing government initiatives promoting digitalization and modernization of public services
- Serve as a potential model for the digital management of grievances and complaints in other barangays

In the long term, this study may contribute to the improvement of grievance and complaint management practices at the grassroots level by demonstrating how digital technologies can enhance documentation, monitoring, record management, and administrative efficiency in local governance.

Storyboard

Log- in Module: designed to manage user authentication and enforce strict role-based permissions

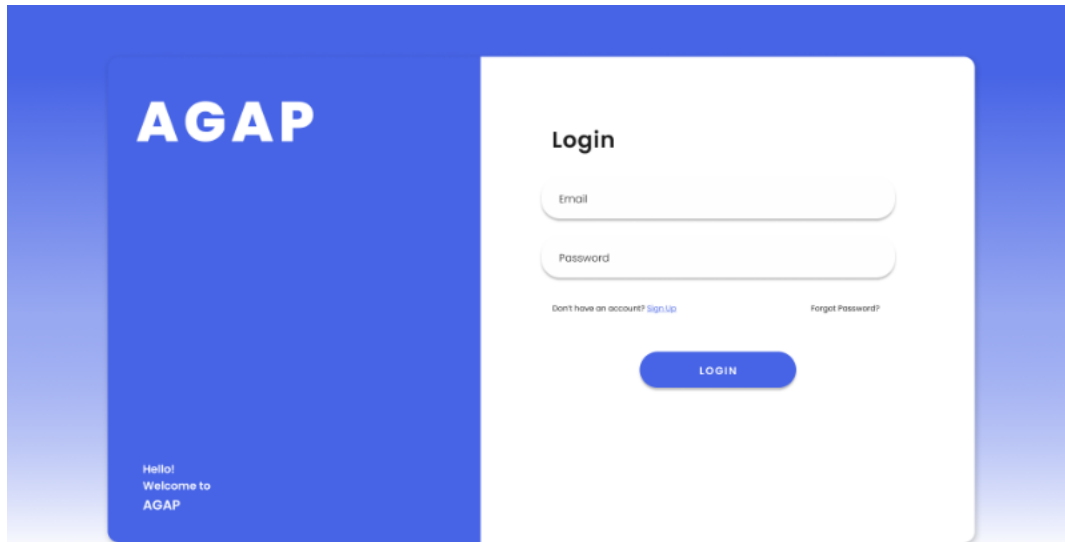


Figure 16.

Dashboard, Monitoring & Reporting System: Provides administrators with a real-time overview of case statuses and mediation progress, while also assisting in generating the mandatory statistical reports required for DILG compliance

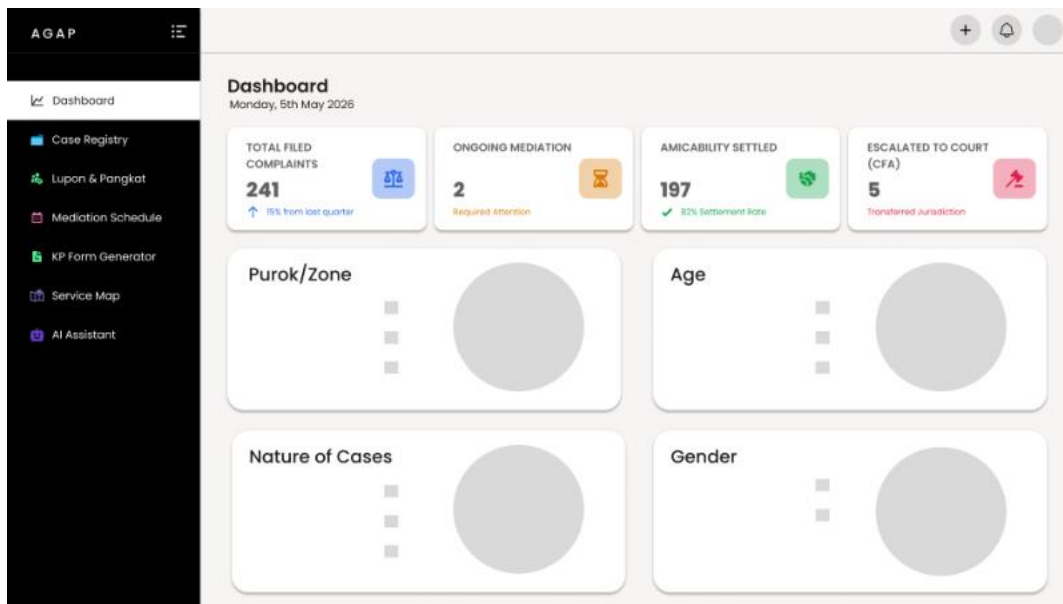


Figure 17.

Search, Retrieval & Record Organization: Organizes resident profiles and historical case data into a searchable digital database, enabling officials to quickly find past records and track recurring or repeat offenders

The screenshot shows the AGAP Case Registry interface. On the left is a dark sidebar with navigation links: Dashboard, Case Registry, Lupon & Pangkat, Mediation Schedule, KP Form Generator, Service Map, and AI Assistant. The main area is titled 'Case Registry' and includes a '+ Add Complaint' button. Below this is a filter section with fields for 'Complainant Surname' (with a placeholder 'e.g. Dela Cruz'), 'Case Type' (dropdown set to 'All'), 'Status' (dropdown set to 'All'), and 'Assigned Lupon' (with a placeholder 'e.g. Dela Cruz'). A 'Reset Filter' button is also present. Below the filters is a table with the following data:

#	Case Details	Parties Involved	Case Type	Mediation Status	Lupon Assigned	Digital Folder
1	CS 02-01-26 Filed On: 2020-01-26	Compr: Santos, Maria Dela Cruz Respr: Reyes, Jose San Juan	Stander	Settled	Jaime Deocarea,	Open Folder

Figure 18.

Case Management System: Serves as the core digital hub to formally record, categorize, and track all grievance cases, complaints, mediation activities, and resolution outcomes from start to finish

The screenshot shows the AGAP Case Registry interface with a detailed view of a case. The sidebar is the same as in Figure 18. The main area is titled 'Case Registry' and includes a '+ Add Complaint' button. Below this is a 'Case Information' section with the following fields:

- Nature of Case
- Complainant
- Respondent
- Address
- Action taken
- Hearing Schedule
- Settlement

On the right side of the 'Case Information' section is a 'Forms' panel with the following items:

- KP Form 7 [Open](#)
- KP Form 8 [Open](#)

Figure 19.

Lupon Assignment & User Management: Secures the platform through role-based access control, ensuring that only authorized personnel can view sensitive dispute records, and manages the proper assignment of cases to specific Lupon members

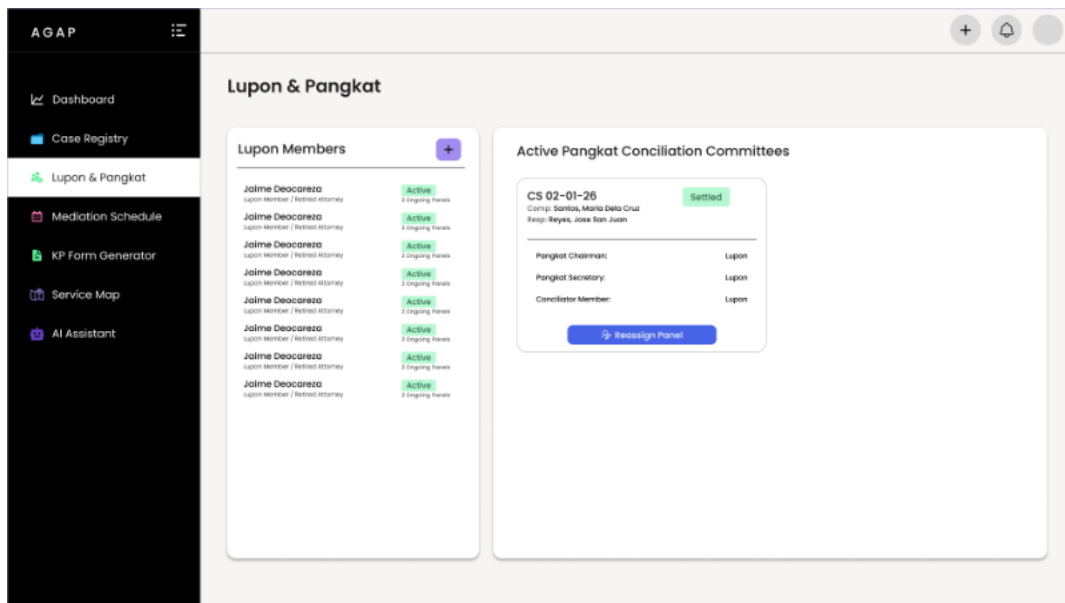


Figure 20.

Scheduling & Hearing Management: Acts as the calendar and notification tool that organizes mediation sessions, tracks hearing dates, manages strict legal deadlines, and alerts users to prevent missed schedules

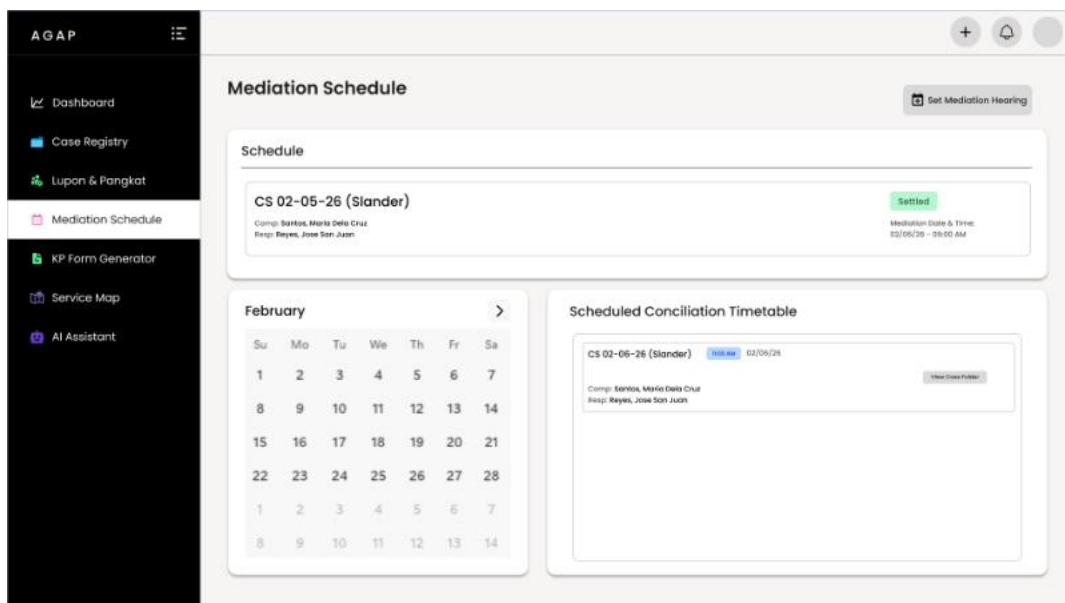


Figure 21.

Automated Document & KP Form Generation: Eliminates repetitive handwriting by automatically producing official Katarungang Pambarangay (KP) paperwork, such as Summons, Notices of Hearing, Settlement Agreements, and Certificates to File Action (CFA)

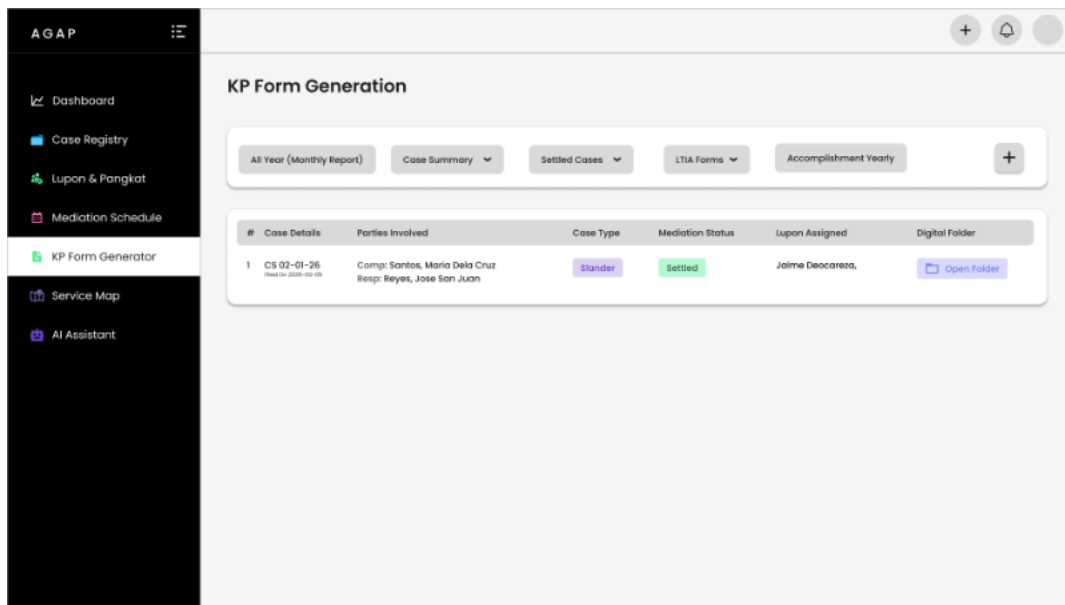


Figure 22.

Location & Proof-of-Service Features: Allows users to attach map data to pinpoint exact incident locations and provides a dedicated space for Summon Servers to officially document their field verifications and successful delivery of notices

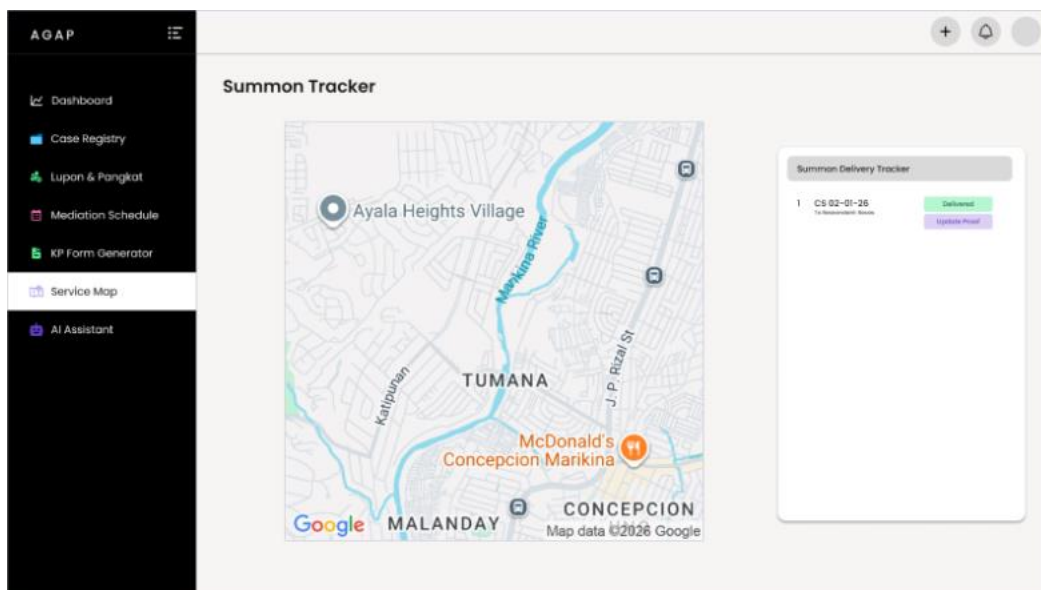


Figure 23.

AI-Assisted Features: Integrates smart tools, including a chatbot to guide residents through filing a complaint and a narrative generator to help Lupon staff convert unstructured verbal statements into formal, organized reports

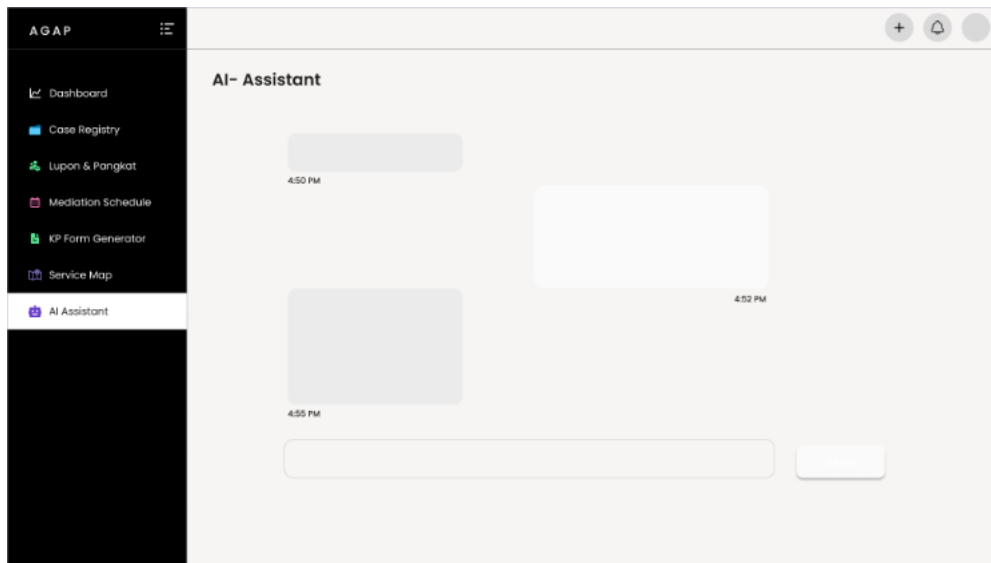


Figure 24.

Who (organizational structure, interview, pamphlet)

Punong Barangay

- Oversees Katarungang Pambarangay operations, receives formal complaints, conducts mediation proceedings, issues notices and summons, constitutes the Pangkat ng Tagapagkasundo when mediation fails, and approves or signs amicable settlements and other official documents.

Barangay Secretary / Lupon Secretary

- Serves as the official custodian of Katarungang Pambarangay records, maintains the KP record book, archives case records, prepares and transmits required documents, and generates monthly and annual reports for monitoring and compliance purposes.

Lupon Clerk

- Assists in the day-to-day administration of Katarungang Pambarangay cases by receiving complaints, encoding case information, preparing KP forms, scheduling hearings, maintaining case records, monitoring case status, retrieving archived documents, and assisting in report preparation.

Pangkat Chairman

- Presides over conciliation proceedings, facilitates discussions between disputing parties, guides the conciliation process, and signs settlement agreements reached during Pangkat proceedings.

Pangkat Secretary

- Records minutes of conciliation proceedings, maintains attendance records, prepares documentation related to hearings, and transmits records and agreements to the Lupon Secretary for filing and archiving.

Pangkat Members

- Participate in conciliation proceedings, assist in resolving disputes, and help facilitate discussions aimed at reaching an amicable settlement.

Summons Server

- Delivers notices, summons, and hearing schedules to complainants, respondents, and other concerned parties, and assists in monitoring service of notices.

Complainant

- Files complaints, provides relevant information and supporting documents, attends mediation and conciliation proceedings, and participates in efforts to resolve disputes.

Respondent

- Receives and responds to complaints, provides necessary information and

supporting documents, attends mediation and conciliation proceedings, and participates in dispute resolution activities.

What

The Lupon Tagapamayapa handled:

Year	Total Cases
2024	325
2025	214

Cases are classified as:

Year	Criminal	Civil
2024	137	188
2025	92	122

Common Complaint Categories

Based on the analysis of Lupon Tagapamayapa case records for 2024 and 2025, the most frequently reported disputes include:

Complaint Category	Description
Non-Payment of Debt	Failure to pay borrowed money, pledged items, utilities, or other financial obligations

Breach of Agreement	Failure to comply with verbal or written agreements, contracts, and settlements
Physical Injuries (Pananakit)	Disputes involving physical harm, assault, or bodily injuries
Defamation (Paninirang Puri / Oral Defamation / Cyber Libel)	Cases involving damage to reputation through spoken, written, or online statements
Threats (Pagbabanta)	Incidents involving intimidation, threats of harm, or coercion
Property and Rental Disputes	Non-payment of rent, eviction issues, property ownership disputes, and damages to property
Disturbance and Public Disorder	Public scandal, harassment, unjust vexation, nuisance, and similar complaints
Malicious Mischief / Property Damage	Intentional destruction or damage to another person's property
Trespassing	Unauthorized entry into private property
Family and Domestic Disputes	Family-related conflicts referred to the Lupon for mediation

Forms used during the current process:

Form	Purpose
KP Form 7	Complaint
KP Form 8	Notice of Hearing

KP Form 9	Summons
KP Form 14	Arbitration Agreement
KP Form 15	Arbitration Award
KP Form 16	Amicable Settlement
KP Form 20–22	Certification to File Action (CFA)

Scope of the Current Workflow

The system will digitally map and automate the following manual procedures currently executed by the Lupon:

1. Complaint filing
2. Case docketing
3. Summons issuance
4. Mediation
5. Pangkat formation
6. Conciliation
7. Arbitration (when applicable)
8. Issuance of Certification to File Action
9. Archiving and record keeping

Proposed System Modules & Data Entities

To address these requirements, the proposed system shall manage and track:

- Complaint records
- Case assignments
- Hearing schedules
- Summons generation
- KP Form generation
- Settlement records
- Arbitration records
- Certification to File Action records
- Case archives

When

The Katarungang Pambarangay process follows prescribed timelines:

Activity	Timeline
Summons Issuance	Next working day after complaint
Initial Hearing	Within 5 days
Mediation Period	15 days
Pangkat Formation	Immediately after failed mediation
Conciliation Period	15 days
Conciliation Extension	Additional 15 days if justified

Arbitration Award	6–15 days from agreement to arbitrate
Settlement Transmission	11–15 days after settlement
Arbitration Transmission	Within 5 days after award
Issuance of CFA	Reasons for Issuance: - Upon failure of conciliation (after 15–30 days) - Upon valid repudiation (within 10 days of settlement) - Upon the respondent's unjustified failure to appear.

The proposed system will assist users in monitoring deadlines, hearing schedules, and case progress.

Where

Current Katarungang Pambarangay activities are conducted in:

- Barangay Hall
- Lupon Office
- Hearing Rooms
- Incident Locations (for verification)
- Records Storage Areas

Physical records are currently maintained through:

- Logbooks
- Filing cabinets
- Printed KP Forms
- Paper-based reports

The proposed system centralizes these records into a web-based platform accessible by authorized personnel.

How

Step 1: Informal Complaint at the Barangay Tanod (BPSO) Level

- Filing the Blotter: The complainant first goes to the desk officer of the Barangay Public Service Office (BPSO) or barangay tanod
- The officer asks for the basic details (who, what, when, where, why, and how) and records the complainant's statement in a blotter
- Checking Jurisdiction: The desk officer verifies if the case is valid for the barangay. It must be person-to-person (not involving corporations), the respondent must reside within the same city or an adjacent barangay, and it must be a "light offense" (penalty not exceeding 1 year imprisonment or a 5,000-peso fine)
- Initial Meeting: The desk officer summons both parties to face each other. If they agree to terms, a settlement is made right there
- If they cannot agree, the case is elevated to the Lupon

Step 2: Preliminary Conversation with LISO

- Screening and Dialogue: Before making it a formal, docketed case, the respondent is invited to the barangay to face the Lupon Information and Screening Officer (LISO)

- The LISO verifies the venue and jurisdiction again and facilitates a "pre-conversation" or preliminary dialogue
- If the case actually belongs to another agency (like cybercrime to the PNP, or traffic issues to the Traffic Management Office), the LISO will refer it appropriately
- If the parties settle during this pre-conversation, the LISO writes up the agreement. If not, the complainant is advised to formally file the case

Step 3: Formal Filing of Complaint

- **Payment and Docketing:** The complainant goes to the clerk, who assigns a case title based on the blotter. The complainant receives an order of payment and pays the filing fee
- The complaint is then given a formal barangay case number (docket number) and scheduled for its first hearing
- Formal summons are sent to both the complainant and the respondent within 5 days

Step 4: Mediation by the Barangay Captain

- **15-Day Mediation:** The Barangay Captain (Punong Barangay) has 15 days to mediate the dispute
- **Settlement or Arbitration:** If successful, they reach an Amicable Settlement
- Alternatively, if both parties agree in writing, they can ask the Captain to decide the case through an Arbitration Award, which becomes final if neither party repudiates (rejects) it within 10 days
- If no settlement is reached within the 15 days, the case is forwarded for the creation of a Pangkat (conciliation panel)

Step 5: Conciliation by the Pangkat

- **Forming the Panel:** Within 3 days of a failed mediation, the Kapitan forms a Pangkat. Both the complainant and respondent must agree on the specific Lupon members who will handle their case
- **15-Day Conciliation:** This panel has 15 days to conciliate the dispute, which can be extended for another 15 days if necessary
- **Issuance of CFA:** If the parties still cannot reach an agreement after the extended conciliation period, the barangay will issue a Certification to File Action (CFA)
- This document allows the complainant to finally escalate the case to the regular courts or the appropriate government office